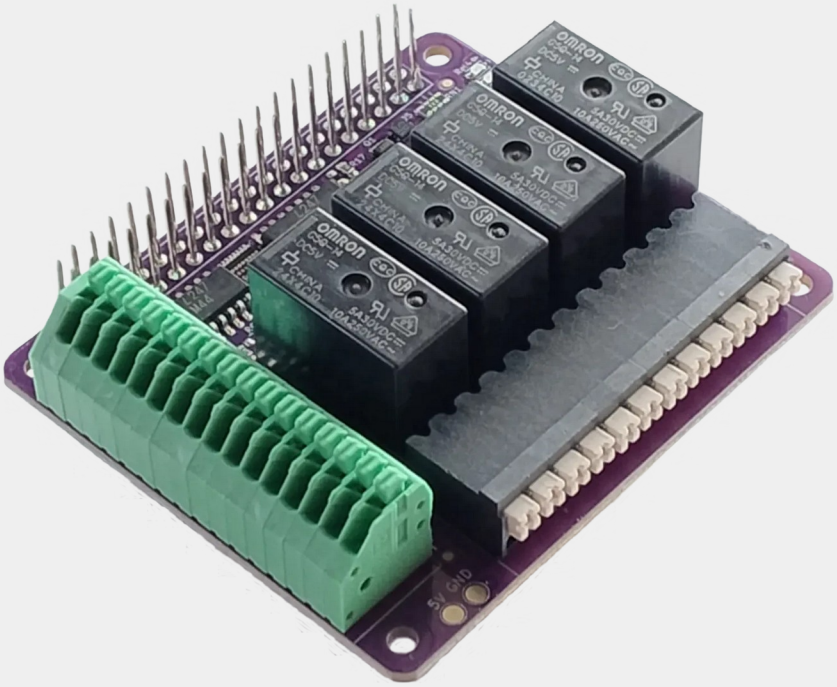
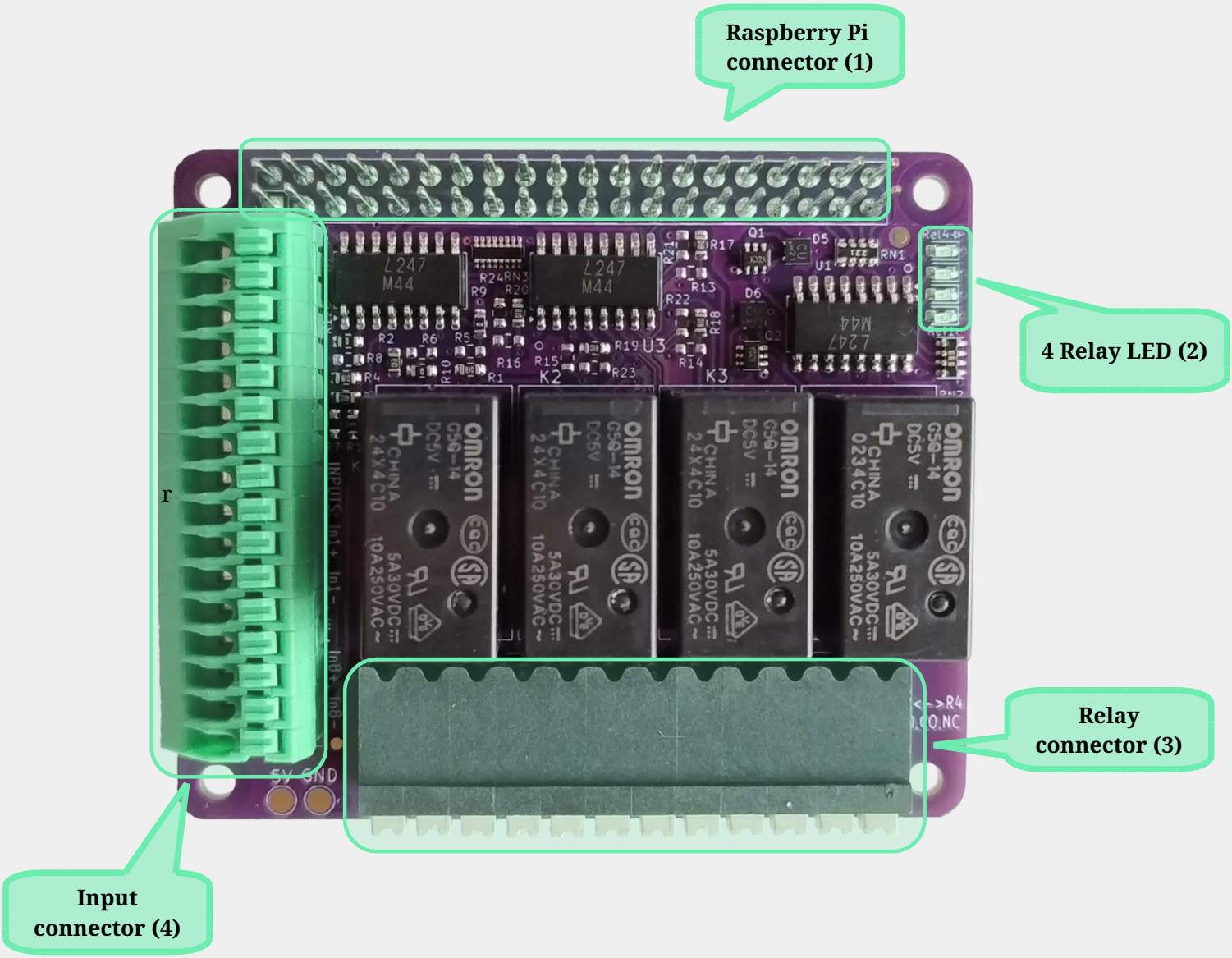


qPIO v1.0

User Manual v1.0
2/9/2025

Please read the instruction manual for safe use and a user experience tailored to your needs.

The qPIO has several connection, setup and feedback points for using it, highlighted in the figure below. For easy identification, the "( )" indicates the subsequent reference numbers. The colour and physical size of the connectors may differ from those shown in the figure below. The points of interest for the qPIO user are shown in the figures below.



1 Safety regulations

1.1 Personal safety

The qPIO has no replaceable parts - only the manufacturer or an accredited service centre can carry out repairs and maintenance.

1.2 Product safety

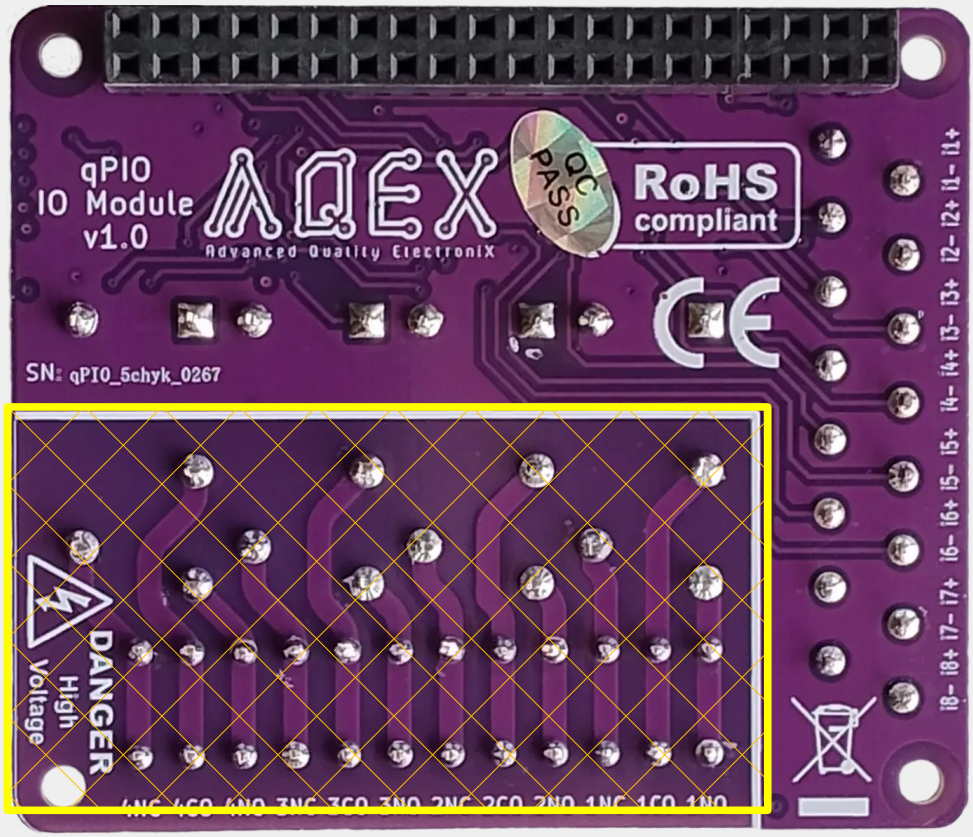
The qPIO product should be protected from too high or too low temperatures, direct sunlight. It should be kept in a dry place for 24 hours before installation.

Conductive liquids, plastic materials may cause short circuiting and permanent product damage, therefore avoid installation in such environments.

1.3 Precautions

The system operates from 5V, which is low voltage, so it is protected from electric shock in a life-safety sense. In case of short circuits caused by foreign matter, the contacts may heat up and cause injury!

The relays are suitable for switching voltages up to 250 VAC. If the device is used to control a high voltage device, touching the relay outputs or the (3) terminals is life-threatening. This section is marked with a yellow outline in the diagram below.



2 Introduction

Thank you for choosing AQEX qPIO to control your electronic devices and sensing the external digital signals!

The qPIO has been carefully and thoroughly designed to provide the most efficient way to ensure smooth operation in a wide range of conditions, to meet a wide range of user requirements.

By default, Raspberry Pi compatible microcomputers can only control devices with 3.3V DC, max. 16mA input current consumption via the GPIO port. With qPIO, they are also capable of switching higher voltages, current loads, and external electronic devices with alternating current (AC).

The GPIO ports of Raspberry Pi-compatible microcomputers can only directly receive 3.3V digital input signals and can only detect contacts (short and open circuit) by activating the internal pull-up and pull-down resistors.

The qPIO module offers four advantages over a direct solution:

- Wider input signal level (3-45V)

- Isolation between input and GPIO port – use of optocoupler

- No need to program internal pull-up and pull-down resistors on contact-type inputs

- In the case of level-type inputs, the input points are completely ground-independent from the qPIO and Raspberry, and there is no ground loop in the system.

3 Technical details

3.1 Inputs

The qPIO module has 8 input ports. Due to optical isolation, overvoltage and voltage spikes arriving at the input do not damage the Raspberry Pi.

Any port can be level or contact type. The port type cannot be changed by the user; it is selected during manufacture.

The qPIO uses 8 GPIO pins on a 2x20 pin connector to monitor the inputs. The assignment is shown in Table 2. The assignment is type-independent.

GPIO BCM / BOARD láb (40 érintkezős tűsor)							
Input 1	Input 2	Input 3	Input 4	Input 5	Input 6	Input 7	Input 8
GPIO27 / Pin 13	GPIO17 / Pin 11	GPIO22 / Pin 15	GPIO05 / Pin 29	GPIO16 / Pin 36	GPIO13 / Pin 33	GPIO12 / Pin 32	GPIO06 / Pin 31

Table 2 – Input ports

When using GPIO pins, make sure to set the pin direction correctly! All ports must be defined as inputs. Incorrect settings may damage the qPIO and/or the Raspberry Pi.

3.1.1 Level type Inputs

In case of a level-type input, the logic signal (High/Low, 1/0, True/False) detected by the GPIO port is determined by the voltage difference between the two corresponding points of the input connector (4). Below 3V, the GPIO port status is Low (0, False), while between 3 and 45V, it is High (1, True). Both points of the input are completely ground-independent, and the external device is not at the same potential as the qPIO and the Raspberry Pi.

Input levels above 45V will damage the input.

The input is polarity sensitive. Reverse polarity with a voltage level above 6V may damage the input. Reverse polarity below 6V will not damage the input, but no signal will be detected on the GPIO pin.

3.1.2 Contact type Inputs

For contact-type ports, a short or open circuit must be connected between the two corresponding points of the input connector row (4) (e.g., relay output). In the case of a short circuit, the corresponding GPIO port detects a Low (0, False) level, and in case of open circuit, it detects a High (1, True) level. The contact-type port is polarity-independent.

Do not connect voltage to the contact type input, as this may damage the external device and/or the qPIO.

3.2 Outputs

The high power is switched by 4 high quality Omron relays, which operate 4 independent circuits in parallel. The relay output acts as a toggle switch. The “COM” pin is the common point, which is connected to the „NC” (Normally Connected) pin in the normal state and to the “NO” (Normally Open) pin when current is applied on the relay input. The ON status of the relays is indicated by one of the LEDs (2).

Opto-isolators between the logic and the relays isolate the 2 systems from each other and the relay input is completely isolated from the output circuit (double isolation).

The maximum voltage that can be switched by the relays under resistive load is 250V for alternating current (AC) and 30V for direct current (DC).

The maximum current that can be switched by the relays is 10A (AC) or 5A (DC) in the NO position and 3A (AC/DC) in the NC position.

The qPIO uses 4 GPIO pins on the 2x20 pin connector for control, as shown in Table 3.

GPIO BCM / BOARD pin (40 pin connector)			
Relay 1	Relay 2	Relay 3	Relay 4
GPIO19 / Pin 35	GPIO26 / Pin 37	GPIO20 / Pin 38	GPIO21 / Pin 40

Table 3 – ports for controlling the ralys

4 Commissioning

The qPIO product is ready for use immediately after preparation and unpacking. For computers using a Raspberry PI compatible 40 pin header, the connection is plug-and-play based, while in other cases the two pin +/- connection and the chosen 8+4 pins can be used (See table 2 & 3).

4.1 Power supply

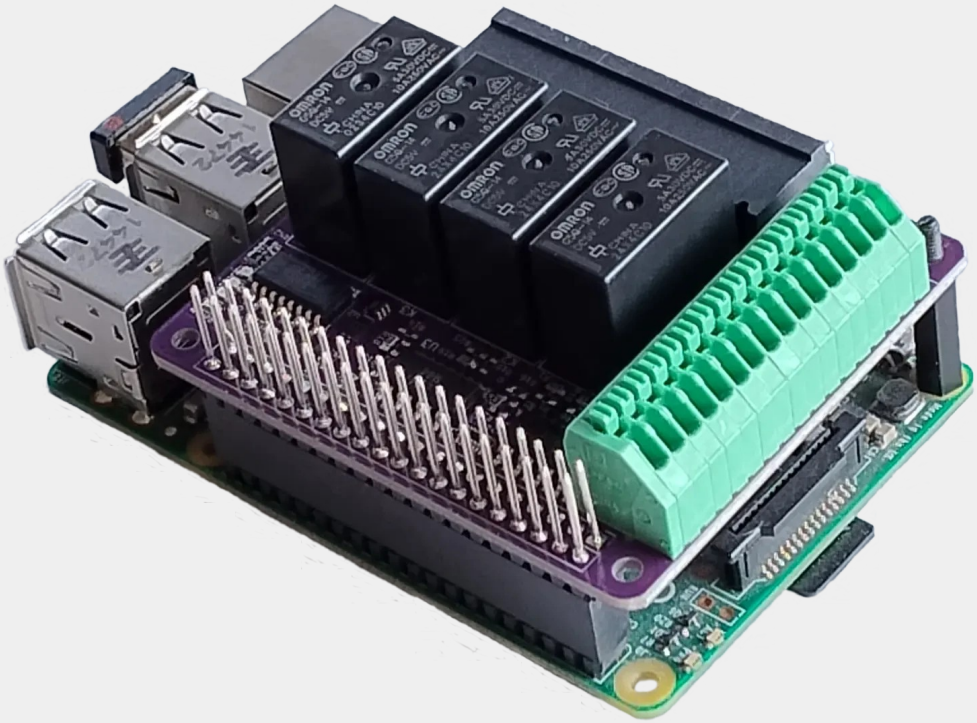
The qPIO is powered by the corresponding 5V legs of the 40 pin connector (1).

4.2 Connections

The controlling device can be connected to the qPIO product in different ways.

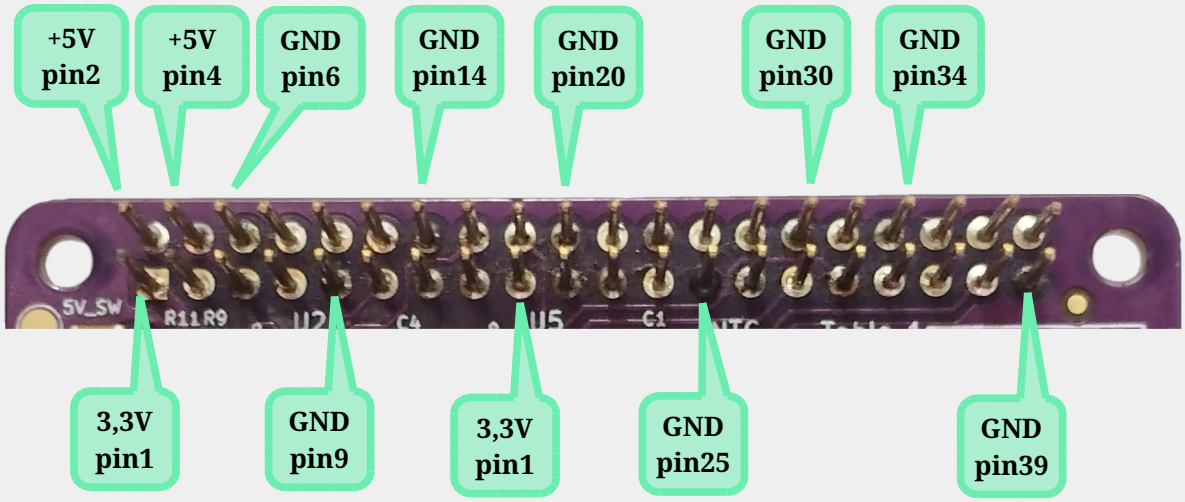
4.2.1 Single board computers (SBC)

If the SBC is equipped with a Raspberry PI compatible 40 pin header, the qPIO product is suitable for HAT connection. Simply plug the qPIO's connector (1) onto the computer's pin header as shown in the picture below.



4.2.2 Other devices

Any device, that can supply 5V DC, can be connected to the qPIO product using the appropriate contacts on the 40-pin header (1).



4.2.3 GPIO pins

The relays can be controlled via the corresponding pins of the connector (1), according to table 3. The inputs are detected at the pins of connector (1) shown in Table 2.

All pins of connector (1) can be accessed on the top side of qPIO. All but the 8+4 GPIO pins are free to be used for any other purpose.

If you connect the qPIO 2x20 pin connector (1) to a device that is not compatible with Raspberry, please note the following:

- The relays can be controlled with 3.3V or 5V logic connected to the selected GPIO pin.

- In case of inputs, the high levels on the corresponding pins of connector (1) will be determined by the voltage connected to the "3.3V" pin (pin1). The maximum value for this is 70V.

4.3 Setup

4.3.1 Hardware setup

The properly connected system will work under normal conditions with default settings.

4.3.2 Software settings

The operating system or program running on the control device can be prepared to switch the relay outputs of the qPIO product via the GPIO connectors corresponding to the configured footprint.

For more information and C and shell based utilities can be found on the https://github.com/aqexhu/qPIO page.

4.4 Troubleshooting

Symptom	Cause of error	Solution